

MENU 3 - SCALE

(POS. 3.1-2) Flow Rate Full Scale 1-2 [FS1-2= l/s xxxx.x] AL2 [FRFS1-FRFS2]

The FS2 (full scale flow rate 2) is active with POS. 6.6 see page 28 ,enabled.

The full scale is used to indicate to the maximum meter's flow rate. The full scale should be chosen carefully as it's parameters are used for several other parameters. There are three fields to fill in order to set this parameter, from left to right: 1) measure unit, 2) time unit of measure and 3) numeric value. The selection is made by positioning the cursor on the field to modify. To change the type unit of measure (metric, Imperial units, mass or volume) see POS. 2.3 see page 27 and POS. 2.19 see page 27 and POS. 2.20 see page 27. The value of Fs1-2 depend nominal diameter POS. 1.4 see page 27. The following tables shown the units of measure available and the conversion factor by comparison with 1dm³ and 1kg. The converter accepts any kind of combination of units of measure satisfying both the following conditions:

❑ Numeric field value 99999

❑ $1/25 \text{ fsmax} \leq \text{numeric field value} \leq \text{fsmax}$.

Where fsmax is the maximum full scale value corresponding to the sensor, equal to a 10m/s liquid speed. The measure units are shown as appear on the display. The Imperial units units are diversified by using capital and small characters.

METRIC	
cm³	Cubic centimeter
ml	Milliliter
l	Liter
dm³	Cubic decimeter
dal	Decalitre
hl	Hectolitre
m³	Cubic meter
ML	Mega Liter

NOT METRIC	
in³	Cubic inch
Gal	American gallon
ft³	Cubic foot
bbl	Standard barrel
BBL	Oil barrel
hf³	Hecto cubic feet
KGL	American Kilo gallon
kf³	Kilo cubic feet
ttG	Ten thousand gallons
IGL	Imperial gallon
IKG	Imperial Kilo gallon
Aft	Acre foot
MGL	Mega gallon
IMG	Imperial mega gallon

MASS UNIT NOT METRIC	
Oz	Ounce
Lb	Pound
Ton	Short tons

MASS UNIT METRIC	
g	Gram
kg	Kilogram
t	Ton

When a measure mass unit is set, the specific gravity function is automatically enabled by the system. Please, note that the mass measure is heavily affected by the temperature. With certain liquids this may cause significant measurement errors. The following measure of time units can be selected: **s** = second, **m** = minute, **h** = hour, **d** = day .

NOTES FOR USING THE MCP INTERFACE

The command FRFS1 =? and command FRS2 = ?, edited by MCP software, return a list of only the unit compatible with the nominal diameter set. If the sensor is insertion type and the diameter is zero, the only possible unit is m/s if the flow rate were chosen metric units, else f/s for the unit of measurement non metric.

(POS. 3.2) *Batching quantity***AL2****[BTQTY]**

This function is enabled when output1 is set to "BATCHING".

It allows to set the unit of measure and the batching quantity.

(POS. 3.3-3.5) *Output Pulse 1-2* [Pls1-2= dm³ x.xxxxx]**AL2****[OP1PV-OP2PV]**

Pls1 and Pls2 is active with POS. 7.1 see page 29 and POS. 7.2 see page 29, enable and setting pulse value on channel 1 and channel 2. This function allows the user to set a signal (a pulse) to be given from the converter when a defined amount of liquid has passed through the sensor. To set the parameter, complete the 2 fields, from left to right: 1) measure unit, 2) numeric value. The selection is performed by positioning the cursor in the field to be modified. To change the unit type (metric, British or American, mass or volume) see POS. 2.4 see page 27 and POS. 2.5 see page 27, POS. 2.19 see page 27 and POS. 2.20 see page 27. The value of Pls1-2 depends on nominal diameter POS. 1.4 see page 27. Only those units described (POS. 3.1 see page 28-POS. 3.3 see page 28) above are available to be selected.

(POS. 3.4-3.6) *Output1-2 Pulse Time* [Tpls1-2= ms x.xxxxx]**AL2****[OP1PT OP2PT]**

Tpls1 and Tpls2 is active with POS. 7.1 see page 29 and POS. 7.2 see page 29 enable. Setting duration of the pulse generated on channel 1 and 2.

With the liquid volume to generate the pulse value (POS. 3.4 see page 28 POS. 3.6 see page 28) set by the user. The user must set the corresponding duration of the pulse to be outputted. This value is expressed in milliseconds and has to be between 0.4 and 9999.99. When the high frequency output is present, then the minimum value can type of device is connected to the converter, the user must verify that the set pulse duration is compatible with the external device processing such pulses. If, for example, an electro-mechanical pulse counter is connected, a minimum pulse time of 0.04 milliseconds can be set.

ATTENTION: The converter can not detect problems that may occur; firstly, the pulse is too long the coils may burn out, secondly, if the pulse is too short, the counter may not be able to function, causing damage of the output.

(POS. 3.7-3.8) *Output full scale freq.1-2***[Frq1-2= x.xxxxx]****AL2****[OU1FF-OU2FF]**

Frq1 and Frq2 are activated with POS. 7.1 see page 29 and POS. 7.2 see page 29 enabled and set to the value freq+/-/+-. Setting duration of the pulse generated on channel 1 and 2.

MENU 4 - MEASURE

(POS. 4.1) Damping

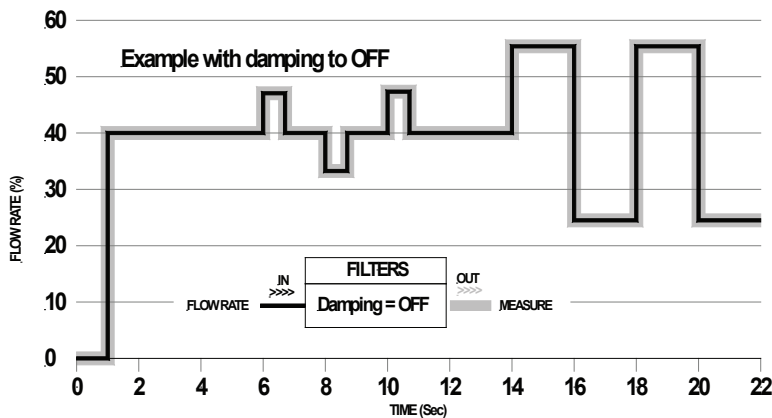
[Damping=OFF/SMART/(TIME)]

AL3

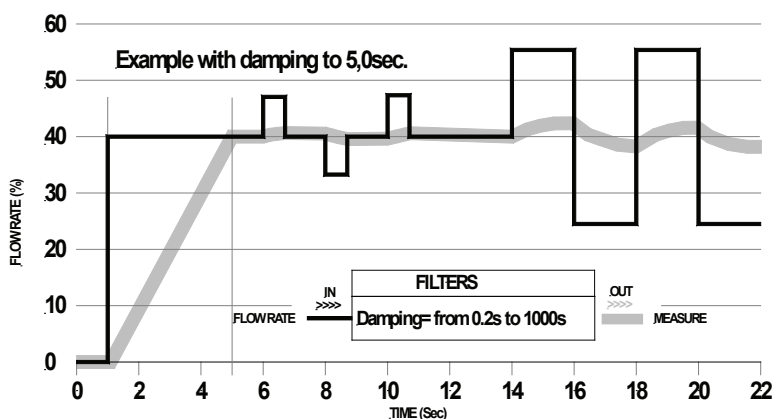
[MFDMP]

This section of manual is extremely important because the correct setting of the filters allows to obtain a proper response of the instrument to the measured flow rate and the specific requirements of use; as a general rule, consider that, starting from Damping = OFF (no filter applied to the measure), successive values, introduce increasing damping. The following diagrams show the instrument's response to changes in flow rate from 0 to 100%, using the different settings of the damping function.

The SMART is an adaptive filter that adapts automatically to most of the processes (recommended value), making the response of the meter very ready to fast changes of flow and at the same time extremely precise and stable for slow variations. It may be convenient to use a constant damping filter time, where there is a pulsating flow (for example generated by peristaltic pumps). With longer times you get a mean value stable, while with short times the measure will closely follow the flow pulses, but consequently more unstable. NOTE: If the rechargeable battery is active, the damping could be set only in "SMART".



Damping function(OFF). the meter follows the trend of fast changes in flow.



Damping mode based on time (fro 0.2s to 1000s) The measure is averaged over a number of samples determined by the value assigned to the dampening function. When the damping parameter is expressed in seconds, the filter works damping the measurement noise and sudden change of flow rate. Increasing the parameter of damping increases the stability of the measurement.

(POS. 4.2) Cut-off threshold**[Cut-off=% xxx]****AL3****[MFCUT]**

Setting the low flow cutoff threshold. This function is useful to avoid that flows close to zero, due to the electrical noises from tiny movements of liquid (due for example to vibrations of the pipe) which cause an increasing of the totalizers. The allowed range for this function is 0-25% of full scale set. For most applications a value between 0.5 and 1% is recommended.

(POS. 4.3) Calibration verify**[Cal.verify=ON]****AL3****[ACAVE]**

This function enables an automatic verification of board's coefficients. As the converter performs continuously a large number of tests, we recommend to use this function only in presence of wide range of temperature. Instead it is NOT recommended to use it when the instrument is used in metering applications (batch).

(POS. 4.4) Automatic scale change enable**[Autorange=ON/OFF]****AL3****[ARNGE]**

Enables the automatic change of scale. The meter may have two different working ranges in order to suit to the variable process conditions. In order to get the best results out of this function it is important range N.2 (Fs2) if enabled is bigger than N.1 (Fs1). When the flow rate increases and reaches the 100% of the full scale 1, then the meter automatically switches to scale 2. When the flow rate decreases again reaching a value on scale 2 equal to the 90% of full scale N.1, then the active scale is 1 again. Allowed values for this parameter: ON/OFF.

Notice: when the autorange is enabled, is not allow to use the manual range change (POS. 6.6 see page 28).

This function does NOT increase the accuracy of the measure; its aim is to increase the resolution of 4/20 mA when the meter work at very low flow rates (typical case the flow rate of water distribution with daytime flow much higher than the night flow). N.B: this function is disabled if the batching function is active

(POS. 4.5) High immunity INPUTS**[H.imm. inp.]****AL4****[HIINP]**

The HIINP function (INPut High Immunity filter) introduces a hardware filter to be used ONLY IN CASE OF ABSOLUTE NECESSITY, when the measure is absolutely unstable or it is NOT possible to make the measure, and every possible attempt to reduce or eliminate the noise do not give a positive result, with particular attention of instrument ground connection. When this function is activated (HIINP = ON) the measure will be influenced by an unavoidable error estimated around 1%.

MENU 4 - MEASURE: ONLY MCP FUNCTIONS**Measure Filter Cut-off Threshold 2****[MCP ONLY]****AL3****[MFCT2]**

Setting the low flow cutoff threshold, it is similar to the function in 4.2. The value of this function is NOT visible on display but only with MCP command.

Dynamic sample analysis**[MCP ONLY]****AL4****[DYNSA]**

Reserved to the service

Dynamic sample time**[MCP ONLY]****AL4****[DYNST]**

Reserved to the service

MENU 5 - ALARMS

(POS. 5.1) *Maximum direct flow rate threshold* **[Max+=XXXXX]** **AL3 [FRAXP]**

Maximum value alarm set for direct flow rate set (expressed in technical units). When the flow rate value exceeds such a threshold, then an alarm message is generated. Setting this parameter to zero disables the alarm generation.

(POS. 5.2) *Maximum reverse flow rate threshold* **[Max-=XXXXX]** **AL3 [FRAXN]**

Maximum value alarm set for reverse flow rate set (expressed in technical units). When the flow rate value exceeds such a threshold, then an alarm message is generated. Setting this parameter to zero disables the alarm generation.

(POS. 5.3) *Minimum direct flow rate threshold* **[Min+=XXXXX]** **AL3 [FRANP]**

Minimum value alarm set for reverse flow rate set (expressed in technical units). When the flow rate value falls below such a threshold, then an alarm message is generated. Setting this parameter to zero disables the alarm generation.

(POS. 5.4) *Minimum reverse flow rate threshold* **[Min-=XXXXX]** **AL3 [FRANN]**

Minimum value alarm set for reverse flow rate set (expressed in technical units). When the flow rate value falls below such a threshold, then an alarm message is generated. Setting this parameter to zero disables the alarm generation.

(POS. 5.5) *Hysteresis* **[Hysteresis=% XX]** **AL3 [ATHYS]**

Hysteresis threshold set for the minimum and maximum flow rate alarms. The value of this parameter is expressed as percentage of the full scale value and may be set from 0 to 25%.

(POS. 5.6) *Current output value in case of failure* **[mA V.alarm =% XXX]** **AL3 [OCACV]**

The output current signal can be specified by the user in case of failure of either, empty pipe, coils interrupted, or ADC error. The signal current is set as a percentage (0 to 125%) of the 0/4-20mA current. 125% corresponds to 24mA and does not depend on the selected range (0-20/4-20mA).

The NAMUR NE43 recommendation asks for a alarms signalling value for the current output lower than 3.6mA (<18%) or bigger than 21mA (>105%). It would then be preferable to set the value of this function at the 10%, so that the current value in case of the a.m. cases would be 2 mA, allowing the following diagnostics:

- ☐ current < 2mA - 5%: line interrupted, power supply failure or faulty converter;
- ☐ 2mA -5% * current * 2mA + 5%: hardware alarm;
- ☐ 4mA * current * 20mA: normal working range;
- 20mA < current * 22mA: out of range, measure above 100% f.s.

(POS. 5.7) *Frequency output value in case of failure* **[Hz V.alarm=%XXX]** **AL3 [OFACV]**

This function is active with POS. 7.1 see page 29 and POS. 7.2 see page 29 enable to (FREQ.+, FREQ.-, FREQ.±) To set the frequency value assigned to the on/off output in one or more of the following failure cases:

- ☐ Empty pipe; Coils interrupted ; ADC error. Allowable range is from 0 to 125% of the frequency full scale value. Although there are no specific rules regulating cases such as these, it would be convenient to use the failure information as follows:
- ☐ 0% Hz * frequency * 100% f.s.: normal working range;
- ☐ 100% f.s. < frequency * 210% f.s.: overflow, measure above the 100% of the f.s.;
- ☐ 115% f.s. * frequency * 125% f.s.: hardware alarm condition.

MENU 6 - INPUTS

WARNING!**THIS MENU IS NOT ENABLED WHEN THE BATCHING FUNCTION IS ACTIVE.****(POS. 6.1-2) *T.direct(pos)flowtot./part reset enable* [T/P+/RESET=ON/OFF] AL3 [VTTPE] [VTPPE]**

When one of this function is enabled, the related totalizer + may be reset through the on/off input.

(POS. 6.3-4) *T.rev.(neg)flowtot./part reset enable* [T/P-/RESET=ON/OFF] AL3 [VTTNE] [VTPNE]

When one of this function is enabled, the related totalizer - may be reset through the on/off input.

(POS. 6.5) *Totalizers counting lock command* [COUNT LOCK= ON/OFF] AL3 [TCLIE]

Totalizers counting lock command enable. When this function is active, applying a voltage on the on/off input terminals the system stops the totalizers no matter which is the flow rate.

(POS. 6.6) *Measure zero lock input enable* [Meas.lock=ON/OFF] AL3 [MSLIE]

When this function is active (ON), applying a voltage on the on input terminals, the measurement is stopped, the meter will display zero flow.

(POS. 6.7) *Calibration Func. input enable* [Calibration=ON/OFF] AL3 [CALIE]

When this function is active, applying a voltage on the on/off input terminals the meter performs a autozero calibration cycle. ATTENTION: If the voltage pulse is less than 1 sec., the meter performs a calibration cycle to compensate possible thermal drifts. If the voltage pulse is more 1 sec, the meter performs a zero calibration measure. To perform the calibration it is absolutely necessary for the sensor to be full of liquid and that the liquid is perfectly still. Even very small movement of the liquid may affect the result of the calibration, and, consequently, the accuracy of the system.

(POS. 6.8) *Flow rate range change inp.* [RANGECHANGE=ON/OFF] AL3 [SRCIE]

Range change external command enable. When this function is enabled, applying a voltage on the on/off input terminals the meter switches to the second measuring range (Fs2).

N.B.: the autorange doesn't allow to use the manual range change see (POS. 4.4 see page 28).

MENU 7 - OUTPUTS

(POS. 7.1) Output 1 function selection**[Out1=XXXXXX]****AL3 [OUT1F]**

Function choice corresponding to digital Output 1. The functions are listed in the table below.

(POS. 7.2) Output 2 function selection**[Out2=XXXXXX]****AL3 [OUT2F]**

Function choice corresponding to digital Output 2. The functions are listed in the table below.

FUNCTIONS FOR OUTPUTS 1 AND 2

- ☐ **OFF:** DISABLE
- ☐ **MAX AL. +:** MAX DIRECT FLOW RATE OUTPUT (ENERGIZED = AL. OFF)
- ☐ **MIN AL. +:** MIN DIRECT FLOW RATE OUTPUT (ENERGIZED = AL. OFF)
- ☐ **MAX/MIN+/-:** MAX/MIN DIRECT FLOW RATE OUTPUT (ENERGIZED = AL. OFF)
- ☐ **MAX AL.-:** MAX INVERSE FLOW RATE OUTPUT (ENERGIZED = AL. OFF)
- ☐ **MIN AL.-:** MIN INVERSE FLOW RATE OUTPUT (ENERGIZED = AL. OFF)
- ☐ **MAX/MIN-:** MAX/MIN INVERSE FLOW RATE OUTPUT (ENERGIZED = AL. OFF)
- ☐ **P.EMPTY:** EMPTY PIPE ALLARM OUTPUT (ENERGIZED = AL. OFF)
- ☐ **HARDWARE AL.:** SUM OF ALL ALARMS "energized interrupted " AND "error input signal "
- ☐ **OVERFLOW:** OUT OF RANGE ALLARM OUTPUT (ENERGIZED = FLOWRATE OK)
- ☐ **ALL ALARMS:** SUM OF ALL ALARMS POSSIBLE
- ☐ **EXT. COMM:** OUTPUT MAY TAKE A STATE DEPENDENT ON AN EXTERNAL CONTROL REMOTELY VIA MCP
- ☐ **FLOW RATE SIGN.:** FLOW DIRECTION (ENERGIZED WHEN FLOW IS NEGATIVE)
- ☐ **SCALE:** INDICATION SCALE
- ☐ **FREQ.+:** FREQUENCY POSITIVE FLOWRATE
- ☐ **FREQ.-:** FREQUENCY NEGATIVE FLOWRATE
- ☐ **FREQ.+/-:** FREQUENCY POSITIVE/NEGATIVE FLOWRATE
- ☐ **PULSES.+:** PULSE POSITIVE FLOW RATE
- ☐ **PULSES.-:** PULSE NEGATIVE FLOW RATE
- ☐ **PULSES+/-:** PULSE NEGATIVE/POSITIVE FLOW RATE
- ☐ **BATCHING:** IT'S POSSIBLE TO SET A QUANTITY OF VOLUME THAT HAVE TO BE DELIVERED DURING AN ACTUATOR (VALVE) OPENING-CLOSING CYCLE

BATCHING

Through this function, it's possible to set the volume quantity that must be delivered, managing the opening and the closing of the valve.

To activate it, follow these instructions:

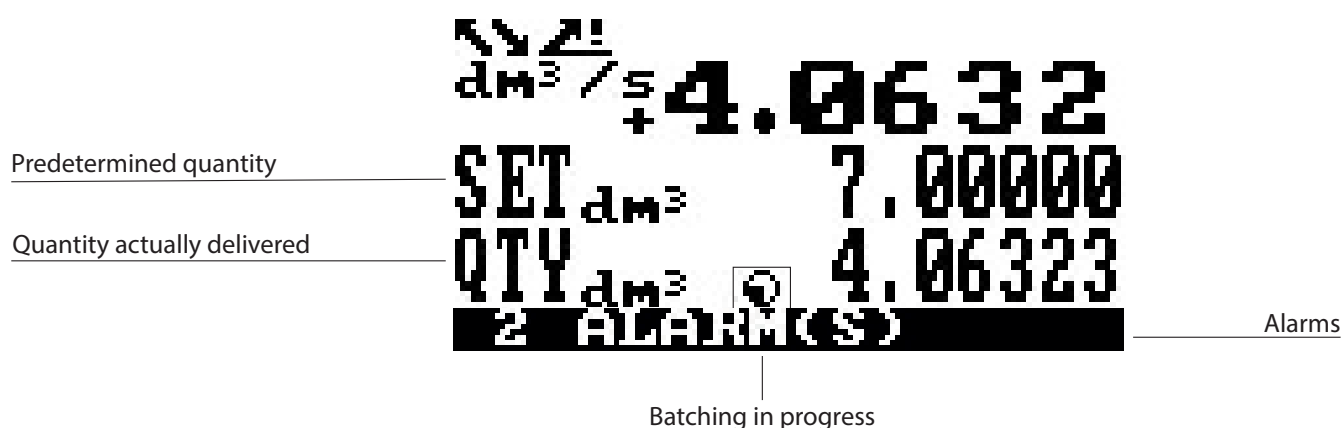
OUTPUTS

```

Out1= BATCHING
Out2= OFF
Out mA1=4_22 +/-
Out mA2=4_22 -0+
A1S=dm³/s 5.0000
A2S=dm³/s 5.0000
t 1 function selectio
    
```

1. Enter to the main menu;
2. Select Output menu;
3. Set the Out1 function on "BATCHING"

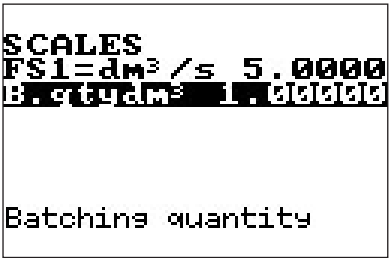
With batching activated, the display visualizes one main page:



The batching parameters settings can be customized according to what the instrument use and application is thought to.

FUNCTION	FUNCTION N° POS.	SETTING
Cut-off	"4.2" pag28	It must be set so that the flow is stably zero when the valve is closed
Damping	"4.1" pag28	It must be set on the minimum value allowed. If T batching ≤ 1s, filters must be disabled

BATCHING QUANTITY SETTINGS:

By display	Enter to the 3. Scale menu; Set the B.qty (batching quantity) value	
By MCP command	"BTQTY?" Through this command it can be set both the unit measure or the quantity (see MCP manual)	VALUES: 0:'ml' 1:'cm3' 2:'l' 3:'dm3' 4:'dal' 5:'hl' 6:'m3' 7:'Ml',0.00001 <> 99999.9
By ModBus interface	Using the ModBus command 16 (see MODbus manual)	

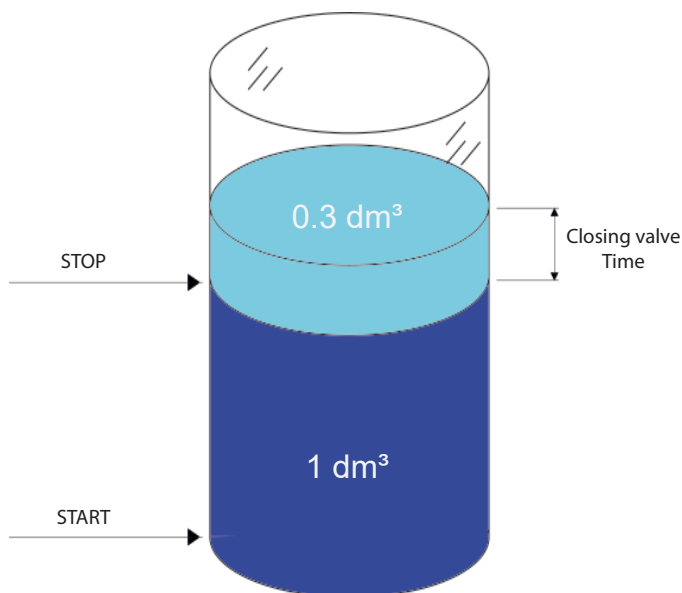
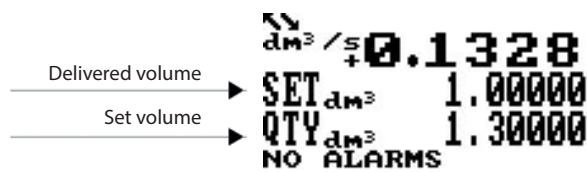
BATCHING OPERATIONS:

ACTION:	OPERATION:	NOTES:
Start/stop by display	Press "UP" key on the keyboard	When the batching process is manually stopped, the OUT1 function is disabled and the batching cycle is interrupted with the valve closing. Every time the batching cycle starts, the set quantity counter and the alarms are reset.
Start/stop by MCP	Type MCP command "BTACT"	
Start/stop by ingresso	External signal connected to digital input 1. N.B: The signal must be a pulse and its duration must be greater than the time of the measurement cycle. See "DIGITAL INPUT ON/OFF OPERATION" page 15	
Start/stop by ModBus	Send the MODbus command 5 (see MODbus manual)	
Visualize batching status by display	Batching in progress symbol on the display 	
Visualize batching status by MCP	Type the MCP command "BTACT?" If = 0 the actuator is closed If = 1 the actuator is open (batching in progress)	

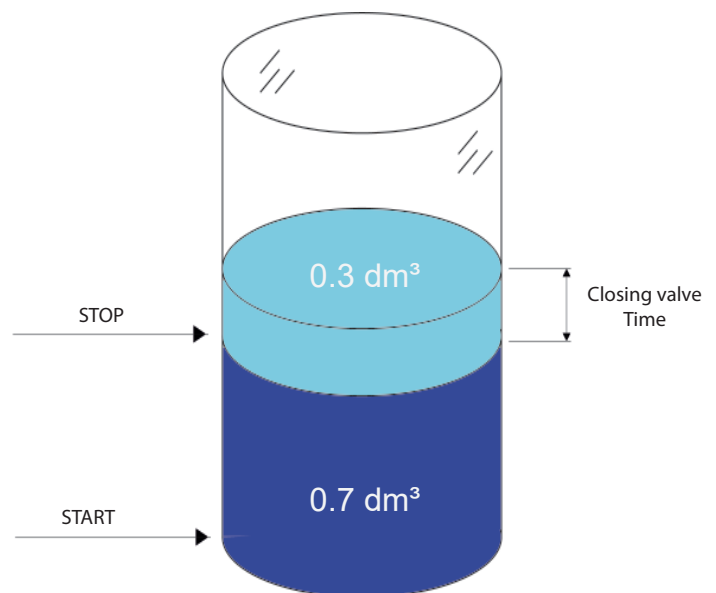
The batching process automatically stops the valve when the batching quantity reaches the set value. If the delivered volume doesn't match to the required one, It is necessary to notice empirically the value to set. In this way It is possible to compensate any "delays" in the process after the stop signal.

EXAMPLE WITH DELIVERED VOLUME = 1 dm³

COMPANSATED BATCHING



NOT COMPASATED BATCHING



BATCHING ALARMS

NAME	DESCRIPTION	NOTES
Batching time-out	It activates after 4 seconds from the start command, when any flow is detected	When alarms activate the actuator close automatically Both alarms can be visualized remotely, setting the usc2 function on "HARD W. AL" or "ALL ALARMS"
Overflow	It activates when the batching quantity is more than 20% of the set one.	

Visualize alarms via display:

The presence of a dosage alarm is highlighted by the message: "ALARM"

Visualize alarms via MCP:

Batching page: The presence of a batching alarm is highlighted by the message: "ALARM"

Alarms page: The presence of a batching alarm is highlighted by the message: "BATCHING ERR."

ATTENTION!!**DISABLED FUNCTIONS WHEN BATCHING IS ACTIVATED**

["1.19" pag27](#)

["4.3" pag28](#)

["4.4" pag28](#)

["9.7" pag29](#)

["9.8" pag29](#)

["9.9" pag29](#)

["12.3" pag30](#)

(POS. 7.3-4) Current output option and range [Out mA1/2=X_XX XXX] AL3 [AO1CF] [AO2CF]

This function sets the current output 1 and 2. This function is optional and will not appear unless the option has been requested. There are three fields to modify for this function:

☐ Scale zero: 4 or 0mA

☐ Full scale: 20 or 22mA

☐ Field: + = positive, - = negative, blank = both, -0+ = central zero scale

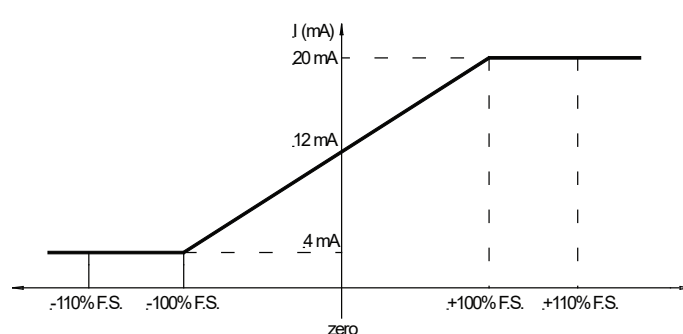
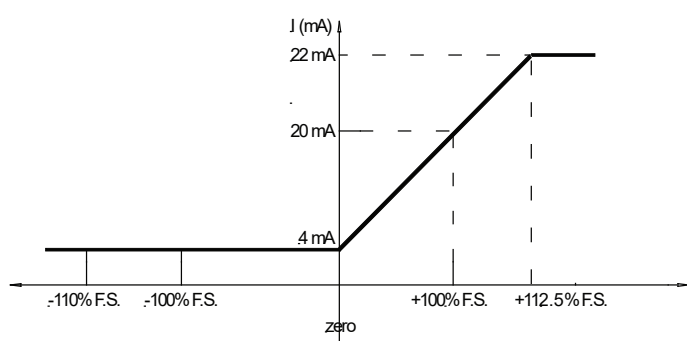
The values corresponding to the scale points are shown in the following chart:

CURRENT VALUES IN mA ASSOCIATE TO THE % FULL SCALE VALUE					
POSSIBLE FIELD	REVERSE FLOW VALUE		ZERO	DIRECT FLOW VALUE	
	≤-210%	-100%		+100%	≥+210%
Out.mA = 0 ÷ 20 +	0	0	0	20	20
Out.mA = 0 ÷ 22 +	0	0	0	20	22
Out.mA = 4 ÷ 20 +	4	4	4	20	20
*Out.mA = 4 ÷ 22 +	4	4	4	20	21.6
Out.mA = 0 ÷ 20 -	20	20	0	0	0
Out.mA = 0 ÷ 22 -	22	20	0	0	0
Out.mA = 4 ÷ 20 -	20	20	4	4	4
Out.mA = 4 ÷ 22 -	21.6	20	4	4	4
Out.mA = 0 ÷ 20	20	20	0	20	20
Out.mA = 0 ÷ 22	22	20	0	20	22
Out.mA = 4 ÷ 20	20	20	4	20	20
Out.mA = 4 ÷ 22	21.6	20	4	20	21.6
Out.mA = 0 ÷ 20 -0+	0	0	10	20	20
Out.mA = 0 ÷ 22 -0+	0	1	11	21	22
**Out.mA = 4 ÷ 20 -0+	4	4	12	20	20
Out.mA = 4 ÷ 22 -0+	2	4	12	20	22

In hardware alarm conditions "HARDW AL." (interrupted coils, empty pipe, measure error) the current value is programmed by the function "mA v.fault" (pos. 5.6) and it is expressed as percentage of a fixed current range, where: 0% = 0mA and 210% = 22mA.

* Example 1: out 4-22 +

** Example 2: out 4-20 -0+



(POS. 7.5) Analog Output1 full scale

[A1S= dm/s x.xxxx]

AL3 [AO1FS]

It allows to set the full scale value for analog output 1 **independently** from the main scale of the instrument.

(POS. 7.6) Analog Output2 full scale

[A2S= dm/s x.xxxx]

AL3 [AO2FS]

It allows to set the full scale value for analog output 2 **independently** from the main scale of the instrument.

MENU 8 - COMMUNICATION

(POS. 8.1) Hart Preamble	[HART pr.=XXXXXX]	AL3	[HARTP]
---------------------------------	--------------------------	------------	----------------

Number of preamble (Hart function)

(POS. 8.2) Hart O.C	[HART O.C=ON/OFF]	AL3	[HRTOC]
----------------------------	--------------------------	------------	----------------

Hart bus output control 4-20mA.



(POS. 8.3) Device Address	[Dev. Addr=XXXXXX]	AL3	[DVADD]
----------------------------------	---------------------------	------------	----------------

Device communication address number.

(POS. 8.4) Modbus Speed	[Speed=XXXXXX]	AL3	[MDBSP]
--------------------------------	-----------------------	------------	----------------

Modbus link speed.

(POS. 8.5) Modbus Parity	[Parity=XXXXXX]	AL3	[MDBPA]
---------------------------------	------------------------	------------	----------------

Modbus link parity.

(POS. 8.6) Modbus Delay	[Delay=XXXXXX]	AL3	[MDBDL]
--------------------------------	-----------------------	------------	----------------

Modbus replay delay.

(POS. 8.7) Modbus chars Timeout	[C. timeout=X]	AL3	[MDBCT]
--	-----------------------	------------	----------------

Maximun delay between chars (frames).

MV210 MODBUS PROTOCOL

☐ RS485 HARDWARE CONNECTION

For the hardware connection see the relative section in this manual and MODBUS manual.

☐ DATA WORD FORMAT

The data bytes travelling in serial form on the communication line are enclosed in words which have a fixed length of 10 bits:

1 START BIT

8 DATA BITS = 1 DATA BYTE

1 STOP BIT

Each word contains one byte of data plus additional bits which serve to synchronies and make the communication safer. These extra bits added automatically in the transmission phase by the transmitter integrated circuit. In the reception phase, the reverse operation is executed by the receiver integrated circuit: the eight data bits are extracted and the others eliminated. These operation are executed entirely on a hardware level. The 8 data bits must be serialised staring from bit 0 (the least significant one).

☐ COMMUNICATION SPEED

The millennium series instruments have 4 communication speeds:

4800 bps

9600 bps

19200 bps

38400 bps

❑ SERIAL PORT SETTINGS

Serial port setting:

Data bits: 8

Parity: Manu < 7-Communication >, function - < Parity >

Stop bits: 1

Flow control: none (no control lines no xon/xoff characters used)

❑ GENERAL DESCRIPTION

All data are sent in group of 16 bits registers. The format used is BIG ENDIAN, MSB byte is sent first, LSB is sent last. When a variable is more than 16 bits in size, it uses two adjacent registers. The totalizer values are expressed as integer numbers. For the correct representation of the value in case there is a fractional part, the decimal dot must be placed in the position specified by the next variable register following the totalized value. All values relative to the flow rate are averaged. The number of samples that compose the average value varies depending on the measure sample rate and the MODBUS reading requests. Example: measure sample rate = 50 Hz, MODBUS reading frequency = 10 Hz, Number of samples used for average calculation = $50/10 = 5$.

Read process variable : COMMAND 04.

ADDRESS	SIZE	TYPE OF DATA	MEANING
0000-0001	2 REGISTERS 32 BITS	FLOAT	full scale flow rate in the unit of measure chosen (as can be seen in the display of the instrument)
0002-0003	2 REGISTERS 32 BIT	FLOAT	flow rate value in percentage
0004-0005	2 REGISTERS 32 BITS	FLOAT	flow rate value in the unit of measure chosen
0006-0007	2 REGISTERS 32 BITS	FLOAT	Flow speed in the unit of measure chosen (m/s or ft/s)
0008-0009	2 REGISTERS 32 BITS	UNSIGNED LONG	Totalizer T+ value
0010	1 REGISTER 8+8 BITS	TWO BYTES	First byte (MSB): number of overflows recorded, second byte (LSB): number of decimal places
0011-0012	2 REGISTERS 32 BITS	UNSIGNED LONG	Totalizer P+ value
0013	1 REGISTER 8+8 BITS	TWO BYTES	First byte (MSB): number of overflows recorded, second byte (LSB): number of decimal places
0014-0015	2 REGISTERS 32 BITS	UNSIGNED LONG	totalizer T- value
0016	1 REGISTER 8+8 BITS	TWO BYTES	First byte (MSB): number of overflows recorded, second byte (LSB): number of decimal places
0017-0018	2 REGISTERS 32 BITS	UNSIGNED LONG	Totalizer P- value
0019	1 REGISTER 8+8 BITS	TWO BYTES	First byte (MSB): number of overflows recorded, second byte (LSB): number of decimal places
0020	1 REGISTER 8+8 BITS	TWO BYTES	first byte (MSB): process flags 1, (LSB): process flags 2
0021	1 REGISTER 16 BITS	UNSIGNED SHORT	Number of measure samples used for to calculate the latest read average value of flow rate
0022	1 REGISTER 16 BITS	UNSIGNED SHORT	Equivalent resistance measured between electrode E1 and the common point, in kilo ohm
0023	1 REGISTER, 16 BITS	UNSIGNED SHORT	equivalent resistance measured between electrode E2 and the common point, in kilo ohm
0024	1 REGISTER, 16 BITS	SIGNED SHORT	voltage measured between electrode E1 and the common point, in millVolts
0025	1 REGISTER, 16 BITS	SIGNED SHORT	voltage measured between electrode E2 and the common point, in millVolts
0026	1 REGISTER, 16 BITS	UNSIGNED SHORT	voltage measured at rechargeable terminals, in milliVolts
0027	1 REGISTER, 16 BITS	UNSIGNED SHORT	residual battery capacity in percentage
0028	1 REGISTER, 16 BITS	SIGNED SHORT	CPU temperature in the unit of measure chosen
0029	1 REGISTER, 16 BITS	SIGNED SHORT	Board temperature T1 the unit of measure chosen
0030	1 REGISTER, 16 BITS	SIGNED SHORT	Board temperature T2 the unit of measure chosen
0031	1 REGISTER, 16 BITS	SIGNED SHORT	Flow sensor coil's temperature the unit of measure chosen
0032	1 REGISTER, 16 BITS	UNSIGNED SHORT	latest sensor test result code
0033	1 REGISTER, 16 BITS	UNSIGNED SHORT	number of alarms currently active

Meaning and value of the process flags 1 returned with the register 0020 (MSB):

- ❑ bit 7 (MSB): flow rate alarm MIN (flow rate below the minimum threshold set)
- ❑ bit 6: flow rate alarm MAX (flow rate over the maximum threshold set)
- ❑ bit 5: flow rate sign (1 = negative)
- ❑ bit 4: flow rate below the cut-off value
- ❑ bit 3: measure range active (0= range 1, 1= range 2)
- ❑ bit 2: flow rate measure reset value status (1= measure is forcibly reset to zero)
- ❑ bit 1: volume counters lock status (1= counters are locked)
- ❑ bit 0 (LSB) : internal use, no meaning

Meaning and value of the process flags 2 returned with the register 0020 (LSB):

- ☐ bit 7 (MSB): flow rate overflow (value greater than full scale)
- ☐ bit 6: pulse channel #2 overflow (frequency greater than maximum possible for the given parameters)
- ☐ bit 5: pulse channel #1 overflow (frequency greater than maximum possible for the given parameters)
- ☐ bit 4: measure signal amplitude out of A/D converter range
- ☐ bit 3: measure signal amplitude out of amplifier capability
- ☐ bit 2: input signal error (out of input chain capability)
- ☐ bit 1: coils excitation error
- ☐ bit 0 (LSB) : pipe empty

Meaning and value of the sensor test flags returned with the register 0032:

- ☐ bit 15 (MSB): resistance at electrode E2 is outside the limits respect to the reference value
- ☐ bit 14 : resistance at electrode E1 is outside the limits respect to the reference value
- ☐ bit 13: coil time B is outside the limits respect to the reference value
- ☐ bit 12: coil time A is outside the limits respect to the reference value
- ☐ bit 11: coil temperature is outside the limits respect to the reference value
- ☐ bit 10: coil leakage current is outside the limit
- ☐ bit 09: coil driver output 2 voltage is out of tolerance during test phase 3
- ☐ bit 08: coil driver output 1 voltage is out of tolerance during test phase 3
- ☐ bit 07: coil driver output 2 voltage is out of tolerance during test phase 2
- ☐ bit 06: coil driver output 1 voltage is out of tolerance during test phase 2
- ☐ bit 05: coil driver output 2 voltage is out of tolerance during test phase 1
- ☐ bit 04: coil driver output 1 voltage is out of tolerance during test phase 1
- ☐ bit 03: coil driver power generator voltage is out of tolerance during test phase 2
- ☐ bit 02: coil driver power generator voltage is out of tolerance during test phase 1
- ☐ bit 01: coil driver power generator value is out of tolerance during test phase 2
- ☐ bit 00 (LSB): coil driver power generator value is out of tolerance during test phase 1

Reset totalizers: COMMAND 05.

ADDRESS	SIZE	TYPE OF DATA	COMMAND VALUE	MEANING
0000	1 REGISTER, 16 BITS	UNSIGNED SHORT	0XFF00 (HEX)	Reset the enabled totalizers (same totalizer enabled for reset from digital input).

Diagnostics: COMMAND 08.

ADDRESS	SIZE	TYPE OF DATA	FUNCTION / VALUE
0000	1 REGISTER, 16 BITS	UNSIGNED SHORT	Return query data
0001	1 REGISTER, 16 BITS	SIGNED SHORT	Restart communication
0004	1 REGISTER, 16 BITS	SIGNED SHORT	Activate listen mode
0010	1 REGISTER, 16 BITS	SIGNED SHORT	Clear counters and Diagnostic Register

ADDRESS	SIZE	TYPE OF DATA	FUNCTION / VALUE
0011	1 REGISTER, 16 BITS	UNSIGNED SHORT	Return Bus Message Count
0012	1 REGISTER, 16 BITS	UNSIGNED SHORT	Return Bus Communication Error Count
0013	1 REGISTER, 16 BITS	UNSIGNED SHORT	Return Bus Exception Error Count with CRC error
0014	1 REGISTER, 16 BITS	UNSIGNED SHORT	Return Server Message Count
0015	1 REGISTER, 16 BITS	UNSIGNED SHORT	Return Server No Response Count
0016	1 REGISTER, 16 BITS	UNSIGNED SHORT	Return Server NAK Count
0017	1 REGISTER, 16 BITS	UNSIGNED SHORT	Return Server Busy Count
0018	1 REGISTER, 16 BITS	UNSIGNED SHORT	Return Bus Character Overrun Count

MENU 9 - DISPLAY

(POS. 9.1) *Language for all msn* [Language= ITA/EN] **AL1** [LLANG]

Choice of the language. There are 8 languages available: **GB** = English, **IT** = Italian, **TR** = Turkish, **PL** = Polish, **DE** = German, **FR** = French, **PT** = Portuguese, **ES** = Spanish.

(POS. 9.2) *Display Contrast* [Contrast= x] **AL1** [DCNTR]

Display contrast set. The contrast can change according to the room temperature. The allowed range is from 0 to 9.

(POS. 9.3) *KeyBoard Timeout Time* [Disp. time=s xxx] **AL1** [KBTMT]

This function set display/keyboard inactivity. The set values are from 020 to 255 second.

(POS. 9.4) *Display refresh Frequency* [D.rate=Hz xx] **AL1** [DISRF]

Frequency of the display data update. This parameter effects only the display layout and not the response time of the meter itself. The possible choices are: 1/2/5/10 Hz.

(POS. 9.5) *Display function number* [Disp.fn= x] **AL2** [DISFN]

This function sets the display of the page making it visible when you start the display. For each display page is associated with a number that corresponds to the position.

(POS. 9.6) *Display function lock* [Disp.lock= x] **AL2** [DLOKE]

This function locks the scrolling of the display pages selected by the setting. POS. 9.5 see page 29 function.

(POS. 9.7) *Partial totalizer* [Part. tot= ON] **AL2** [PTOTE]

This function enables the display of partial totalizer in visualization pages

(POS. 9.8) *Negative totalizer* [Neg. tot= ON] **AL2** [NEGTE]

This function enables the display of negative totalizer in visualization pages

(POS. 9.9) *Net totalizer* [Net. tot= ON] **AL2** [NVTTE]

This function enables the display of net totalizer in visualization pages

(POS. 9.10) *Date and Time* [Disp.Date= ON] **AL2** [DATDE]

This function enables the display of date and time in visualization pages

(POS. 9.11) *Quick start menu* [Quick start= ON] **AL2** [QSTME]

This function enables the quick start menu.

MENU 10 - DATA LOGGER

(POS. 10.1) <i>Data logger enable</i>	[D.logger en= /OFF]	AL3	[DLOGE]
--	----------------------------	------------	----------------

This function enables data logger.

The following functions are activated by [D.logger en= ON]

(POS. 10.2) <i>Unite of Measure</i>	[Meas. units= ON]	AL3	[DLUME]
--	--------------------------	------------	----------------

Measure unit recording enable

(POS. 10.3) <i>Field separator character</i>	[Field separ.= ;]	AL3	[DLFSC]
---	--------------------------	------------	----------------

This function will set the separator character between data logger data.

(POS. 10.4) <i>Decimal Separator Char.</i>	[Decim.separ.= .]	AL3	[DLDSC]
---	--------------------------	------------	----------------

This function will set the separator character between data logger number value.

(POS. 10.5) <i>Sample Interval</i>	[Interv.= xx:xx:xx]	AL3	[DLGSI]
---	----------------------------	------------	----------------

Sampling interval. This function set the log frequency. [Interv.= Hours : Minutes: Seconds]

(POS. 10.6) <i>Total direct totalizer</i>	[Log T+= ON]	AL3	[DTTPE]
--	---------------------	------------	----------------

Enable logging of total direct totalizer.

(POS. 10.7) <i>Partial direct totalizer</i>	[Log P+= ON]	AL3	[DTPPE]
--	---------------------	------------	----------------

Enable logging of partial direct totalizer.

(POS. 10.8) <i>Total reverse totalizer</i>	[Log T-= ON]	AL3	[DTTNE]
---	---------------------	------------	----------------

Enable logging of total reverse totalizer

(POS. 10.9) <i>Partial reverse totalizer</i>	[Log P-= ON]	AL3	[DTPNE]
---	---------------------	------------	----------------

Enable logging of partial reverse totalizer

(POS. 10.10) <i>Total Net totalizer</i>	[Log TN= ON]	AL3	[DLTNE]
--	---------------------	------------	----------------

Enable logging of total net totalizer

(POS. 10.11) <i>Partial Net totalizer</i>	[Log PN= ON]	AL3	[DLPNE]
--	---------------------	------------	----------------

Enable logging of partial net totalizer

(POS. 10.12) <i>Log Flow rate in measure unit</i>	[Log Q(UM)= ON]	AL3	[DFTUE]
--	------------------------	------------	----------------

Enable logging of flow rate in measure unit

(POS. 10.13) <i>Log Flow rate in percentage</i>	[Log Q(%)= ON]	AL3	[DFPCE]
--	-----------------------	------------	----------------

Enable recording of the flow rate as a percentage of full scale value set.

(POS. 10.14) <i>Alarm events</i>	[Log AL.EV= ON]	AL3	[DALEE]
---	------------------------	------------	----------------

Enable logging of alarm events

(POS. 10.15) <i>Sensor test result</i>	[Log STR= ON]	AL6	[DSTRE]
---	----------------------	------------	----------------

Enable logging of sensor test results

(POS. 10.16) <i>Board temperatures</i>	[Log BTS= ON]	AL6	[DBTSE]
---	----------------------	------------	----------------

Enable logging of board temperature

(POS. 10.17) <i>Internal board voltages</i>	[Log IBV= ON]	AL6	[DIBVE]
--	----------------------	------------	----------------

Enable logging of internal board voltage

(POS. 10.18) <i>Electrodes DC voltages</i>	[Log EDC= ON]	AL6	[DEDVE]
---	----------------------	------------	----------------

Enable logging of electrodes DC voltage

(POS. 10.19) <i>Electrodes AC voltages</i>	[Log AEC= ON]	AL6	[DEAVE]
---	----------------------	------------	----------------

Enable logging of electrodes AC voltage

(POS. 10.20) <i>Electrodes source impedance</i>	[Log EIZ= ON]	AL6	[DESIE]
--	----------------------	------------	----------------

Enable logging of electrodes impedance

(POS. 10.21) <i>Sensor coils values</i>	[Log SCV= ON]	AL6	[DSCVE]
--	----------------------	------------	----------------

Enable logging of sensor coils value

MENU 10 - DATA LOGGER: ONLY MCP FUNCTIONS

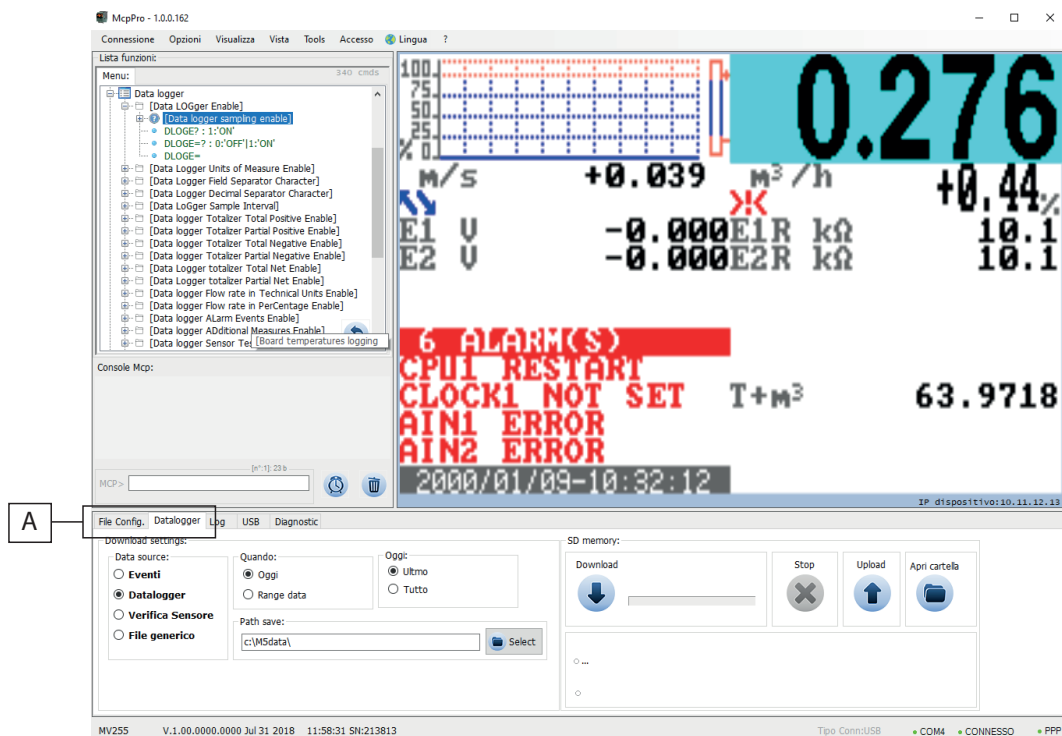
<i>LoG All Information Enable</i>	[MCP ONLY]	AL6	[LGAIE]
--	-------------------	------------	----------------

Log all events information. This function save in the event file all MCP commands.

USING DATA LOGGER BY MCP INTERFACE

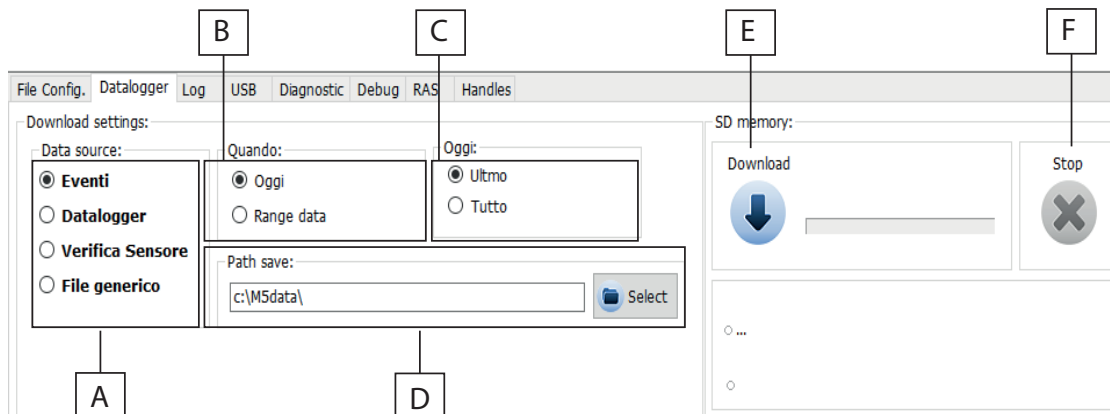
Data are stored on micro SD card; the organization is based on "tree-structure": the system create a daily folder where it save events and data logger . The data can be downloaded by MCP interface.

MCP INTERFACE



Click tab-control data logger to view files.

The sampling data backup depends on the value set by the POS function 10.4 page 25.



A=Data source

Events: Save the file system events (Example F-RAM hardware data [WORKING AREA] [SUCCESSFULLY LOADED])

Data logger: Save files of the enabled data logger function.

Sensor Verify: data logged by BIV function

Generic file:

B=When

Today; It indicates the download file for the current day

Data range; this option allows you to select the date range for download.

C=Today

Last; this option allows to download the latest files, recorder after the last download
All; this option allows the download of all the current day of the file

D=Save path:

This option allows you to save files to the folder on your PC

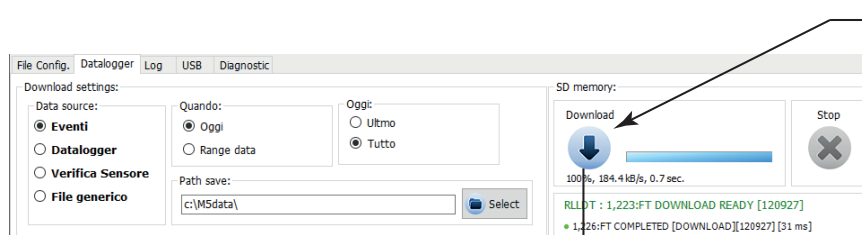
E=Download:

Button to start the download process

F=Stop:

Button to stop the download process

Example: Download Events



Per scaricare tutti gli eventi del giorno odierno in una cartella specifica, settare i seguenti parametri come segue:

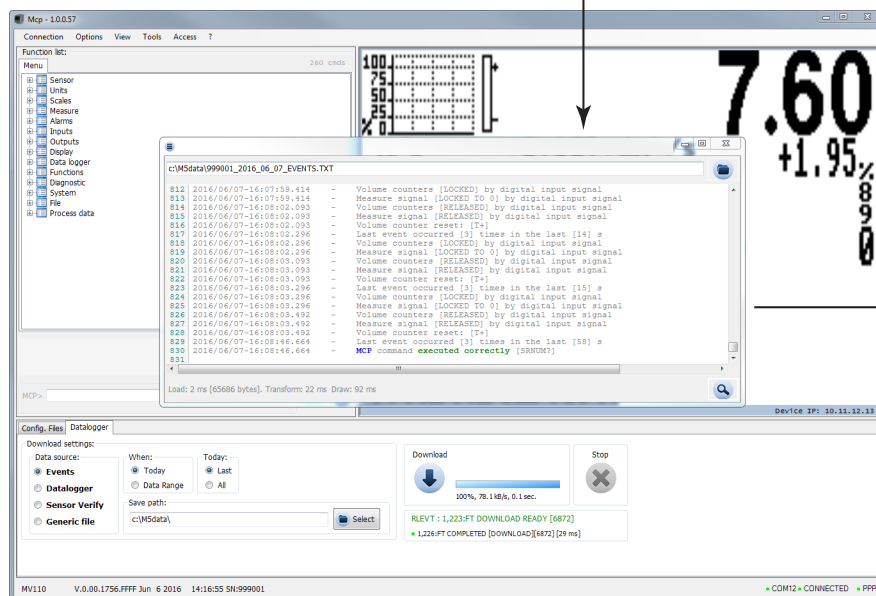
Data source: Eventi

Quando: Oggi

Oggi: Tutti

Save path: C:/.....

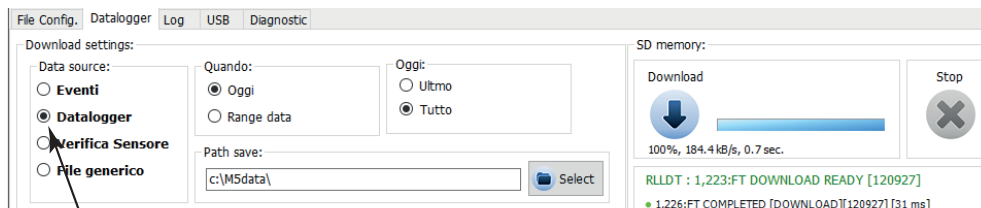
Impostati i parametri cliccare il tasto Download.



Terminato il Download, viene visualizzata una finestra contenente la lista degli eventi in data odierna. Il file viene salvato in formato txt nella cartella specificata per il download.

Example: Download Data Logger

Note: it is recommended the date synchronization between converter and PC to perform correctly the events and logger reading operations.



Per scaricare tutti i dati del giorno odierno in una cartella specifica, settare i seguenti parametri come segue:

Data source: Datalogger

Quando: Oggi

Oggi: Tutti

Save path: C:/.....

Impostati i parametri premere il tasto Download.

View downloaded files setting download data logger.

Note: The fields are in a fixed position, regardless if the above fields are active or not. The disabled fields are empty (delimited by the separator but without data).

N°Record. View progressively the number of registered records.	A NRECORD n°
Date. The recording date viewing for each record.	B DATE ddmm/yy
Hours. Time recording viewing for each record.	C HOURS 000000
Total positive totalizer value. Form Fields when the send flag is active on the totalizer T+.	D U.M. dm3 E T+ 0
Partial positive totalizer value. Form Fields when the send flag is active on the totalizer P-.	F U.M. dm3 G P- 0
Total negative totalizer value. Form Fields when the send flag is active on the totalizer T-.	H U.M. dm3 I T- 0
Partial negative totalizer value. Form Fields when the send flag is active on the totalizer P-.	J U.M. dm3 K P- 0
Total net totalizer value. Form Fields when the send flag is active on the totalizer TN.	L U.M. dm3 M TN 0
Partial net totalizer value. Form Fields when the send flag is active on the totalizer PN.	N U.M. dm3 O PN 0
Flow rate. Form Fields present when the send flag is on the flow in units of measurement.	P U.M. FLOWRATE dm3/s Q 0
Flow rate %. Form fields present when the flag of alarm recording is active (only N ° of present total alarms)	R U.M. FLOWRATE % % S 0
N ° active alarms. Form fields present when the flag of alarm recording is active (only N ° of present total alarms)	T U.M. N°ACTIVE ALARM AL U 0
Loss of current measured during insulation test. Available value when recording the sensor test data is active.	V U.M. OPTI U.M. mA W 0
Time rise A. Available value when recording the sensor test data is active.	X U.M. TRISE A ms Y 0
Time rise B. Available value when recording the sensor test data is active.	Z U.M. TRISE B ms AA 0
Sensor test error code. Available value when recording the sensor test data is active.	AB U.M. CETS ERR AC 0

Visualization of downloaded file. Access Level 2 (diagnostic level) is required in order to download this type of file.

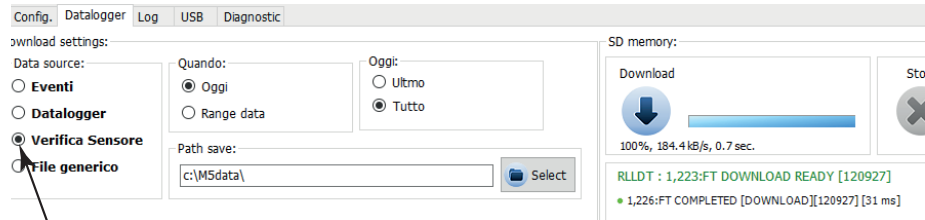
Voltage measured on electrode E1. Form fields when is active the recording of data on the input voltage (diagnostic value).	AD U.M. V 0	AE U.M. V 0
Voltage measured on electrode E2. Form fields when is active the recording of data on the input voltage (diagnostic value).	AF U.M. V -0.023	AG U.M. V -0.023
Differential voltage between the two electrodes. Form fields when is active the recording of data on the input voltage (diagnostic value)	AH U.M. V 0	AI U.M. V 0
Common mode voltage in the electrodes. Form fields when is active the recording of data on the input voltage (diagnostic value).	AJ U.M. V 0	AK U.M. V 0
Noise at low frequency measured on the electrodes. Form fields when is active the recording of data on the input signal noise levels (diagnostic value).	AL U.M. V 0	AM U.M. V 0
Differential low frequency noise measured on the electrodes. Form fields when is active the recording of data on the input signal noise levels (diagnostic values).	AN U.M. V 0	AO U.M. V 0
Low-frequency noise measured input ADC. Form fields when is active the recording of data on the input signal noise levels (diagnostic values).	AP mV 0	AQ U.M. mV 0
High frequency noise measured input ADC. Form fields when is active the recording of data on the input signal noise levels (diagnostic values).	AR mV 0	AS U.M. mV 0
Measured equivalent resistance on the electrode 1. Form fields when is active the recording of data on the electrode resistance measurements (diagnostic values).	AT U.M. kohm 0	AU U.M. kohm 0
Measured equivalent resistance on the electrode 2. Form fields when is active the recording of data on the electrode resistance measurements (diagnostic values).	AV U.M. kohm 0	AW U.M. kohm 0
Coils excitation current. Form fields when is active the recording of data related to the sensor excitation circuit measures (diagnostic value)	AX mA 0	AY U.M. mA 0
Measured resistance of the excitation circuit (coil + cable). Form fields when is active the recording of data relative to the sensor excitation circuit measures (diagnostic values).	AZ ohm 0	BA U.M. ohm 0
Temperature measured on the sensor coils (indirect measurement). Form fields when the data transmission flag is active relative to the sensor excitation circuit measures (diagnostic values).	BB °C 0	BC U.M. °C 0
Temperature T1 (sheet sensor 1). Form fields when the data transmission flag on board the internal temperature measurement is active (diagnostic values).	BD °C 0	BE U.M. °C 0
Temperature T2 (sheet sensor 2). Form fields when the data transmission flag on board the internal temperature measurement is active (diagnostic values).	BF °C 0	BG U.M. °C 0
CPU temperature. Form fields when the data on the card's internal power supply voltage measurements flag is ON (diagnostic value).	BH °C 0	BI U.M. °C 0
Primary power supply of CPU. Form fields when the data on the card's internal power supply voltage measurements flag is ON (diagnostic value).	BJ V 0	BK U.M. V 0
Positive supply voltage of analog circuits. Form fields when the data on the card's internal power supply voltage measurements flag is ON (diagnostic values).	BL V 0	BM U.M. V 0
Negative supply voltage of the analog circuits. Form fields when the data on the card's internal power supply voltage measurements flag is ON (diagnostic values).	BN V 0	BO U.M. V 0
Voltage measured on the battery B1 (NOT rechargeable battery). Fields when the data on the card's internal power supply voltage measurements flag is ON (diagnostic values).	BP V 0	BQ U.M. V 0
Voltage measured on the battery B2 (or rechargeable battery). Form fields when the data on the card's internal power supply voltage measurements flag is ON (diagnostic values).	BR V 0	BS U.M. V 0
% Battery charge. Form fields when the data on the card's internal power supply voltage measurements flag is ON (diagnostic values).	BT % 0	BU U.M. % 0

Example: Sensor Verify

The function "Sensor Verify" creates the "STESTLOG.CSV" file, according to the following conditions:

- 1) Activate the SDC / RTC option in group "HW Config" (by factory see the order code)
- 2) Activate the "BIV" in the group "PRODUCT CODE" (by factory see the order code)
- 3) Activate the "Sens.verify" function in the "Sensor" menu

Activated such conditions, the converter will perform every hour a measure of sensor's parameters and record a line of the file "STESTLOG.CSV"; it is possible even a manual verification by the command "sens.verify" on the menu "Diagnostic" or through the MCP command "SVERC".



Per scaricare tutte le verifiche sensore del giorno odierno in una cartella specifica, settare i seguenti parametri come segue:

Data source: Verifica sensore

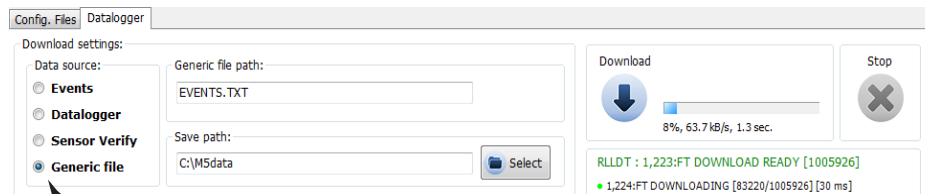
Quando: Oggi

Oggi: tutti

Save path: C:/.....

Impostati i parametri premere il tasto Download.

Example: Generic File



Questa Funzione permette di selezionare un file contenuto nella memoria SD del convertitore e scaricarlo.

Nella sezione "Generic file path" indicare il nome del file contenuto nella SD e indicare in "Save path" il percorso di salvataggio del file stesso. Impostati tali valori procedere con il download del file cliccando sul tasto "download".

MENU 11 - FUNCTION

The following functions are activated by first pressing the "ENTER" and then the "ESC" when the screen appears "confirm" to start the function.

(POS. 11.1) <i>Total direct totalizer reset</i>	[T+ RESET= ON]	AL3	[VTTPR]
--	-----------------------	------------	----------------

Reset total direct totalizer for direct flow rate (+)

(POS. 11.2) <i>Partial direct totalizer reset</i>	[P+ RESET= ON]	AL3	[VTPPR]
--	-----------------------	------------	----------------

Reset total partial totalizer for direct flow rate (+)

(POS. 11.3) <i>Total reverse totalizer reset</i>	[T- RESET= ON]	AL3	[VTTNR]
---	-----------------------	------------	----------------

Reset total reverse totalizer for direct flow rate (-)

(POS. 11.4) <i>Partial reverse totalizer reset</i>	[P- RESET= ON]	AL3	[VTPNR]
---	-----------------------	------------	----------------

Reset partial reverse totalizer for direct flow rate (-)

(POS. 11.5) <i>Load factory default sensor</i>	[Load sens.f.def= ON]	AL3	[LFDSD]
---	------------------------------	------------	----------------

This function resets the parameters of the sensor factory default. To Load the saved files see function (11.7).

(POS. 11.6) <i>Load factory default converter</i>	[Load conv.f.def= ON]	AL3	[LFDSD]
--	------------------------------	------------	----------------

This function resets the parameters of the converter factory default. To Load the saved files see function (11.8).

(POS. 11.7) <i>Save sensor factory default</i>	[Save sens.f.def= ON]	AL6	[SFDSD]
---	------------------------------	------------	----------------

This function save the parameters of the sensor factory default.

(POS. 11.8) <i>Save convert factory default</i>	[Save conv.f.def= ON]	AL6	[SFDSD]
--	------------------------------	------------	----------------

This function loads the data from a converter to another.

(POS. 11.9) <i>Calibration Immediate</i>	[Calibration]	AL5	[CALIC]
---	----------------------	------------	----------------

Perform manually a board's calibration. Press Enter and the message "EXECUTE?" will be visualized on the display then press long the key Enter to proceed. Press any other key to delete the operation.

If the sensor table is valid, the calibration is performed also when one of the following parameter has been change:

1. SENSOR DIAMETER -> Menu Sensor1
2. SENSOR MODEL -> Menu Sensor1
3. Exc. CURRENT -> Menu Sensor1
4. S. Freq. -> Menu Sensor1

To check the calibration status, active or inactive, type the command MCP Calic? and check as follows:

- ☐ CALIC = 1 calibration in progress
- ☐ CALIC = 0 calibration terminated

MENU 11 - FUNCTION: ONLY MCP FUNCTIONS

<i>Save sensor reference data</i>	[MCP ONLY]	AL4	[SRFDS]
--	-------------------	------------	----------------

Save conv.f.def= ON. "meter data" page 75

MENU 12 - DIAGNOSTIC

(POS. 12.1) *Self Test Diagnostic*

[Self Test]

AL3

[ATSIC]

Meter auto-test function. This function stops the normal functions of the meter and performs a complete test cycle on the measure input circuits and on the excitation generator. To activate this function, after select it, push key Enter, at the question: "CONFIRM EXEC.?" Long Push the same key to start auto-test, or any other key for delete operation. At the end of operation the converter will revert to one of the initial visualization pages. This function is automatically performed when switching on the device. This function restarts the converter.

(POS. 12.2) *Test display*

[Test display]

AL1 NO MCP COMMAND

This function allows to do a physical test the graphic display. During this operation, 4 sequences are displayed to test the correct functioning of the device.

(POS. 12.3) *Sensor verify*

[Sens. verify]

AL3

[SVERC]

This function perform a manual sensor verification (if BIV is active)

(POS. 12.4) *Flow rate simulation*

[Flow sim=ON]

AL3

[MSIEN]

Flow rate simulation enabling. With this function it is possible to generate an internal signal that simulates the flow rate, allowing the outputs and all the connected instruments test. After enabling it, a '▲▼' appears in the top left of the screen and the flow rate simulation can be:

- ☐ set: by pushing the key Enter from one of visualization pages, to set the required % flow rate (Fl. rate=%) and the same key to confirm the value;
- ☐ finished: by pushing the key Enter from visualization pages and then by long pushing the same key.

(POS. 12.5) *Diagnostic measures*

[Display measures]

AL5

DMVLS

This Function shows the values of the various internal parameters as listed below :

```

UCPU:V      5.01  LFN_COM:V      0.0000
UPS:V       5.41  LFN_DIF:V      0.0000
VUSB:V      4.53  HFN_ADC:V      0.0000
+AVCC:V     +10.1 LFN_ADC:V      0.0001
-AVCC:V      -9.9 MEAS_NB:V      0.0000
VBATT:V      0.000 CAL_I:V      16.503
IBATT:V      0.000 CAL_U:V      3230.7783
VIN1:V      -8.071 CAL_G:V      0.1633
VIN2:V      -8.065 CAL_O:V      -0.0055
VIN_C:V     -8.068 CAL_C:V      1.0000
VIN_D:V     -0.006 CAL_R0:V      1.0000
C_C:MA      25.00  CAL_R1:V      1.0000
C_U:V        5.0   CAL_R2:V      1.0000
C_UPK:V      5.5   CAL_R3:V      1.0000
C_R:Q       201.9  SVS_F:MHz      50.0051
C_PWR:W      0.126 CURR_K:V      0.0000
C_T:°C       0.00  PROC1:V      46.4
C_R1:MS      0.00  PROC2:V      7.3
C_LK:MA      0.000 PROC3:V      3.8
C-ST:        1     PROC4:V      29.8
S_UER_RSLT:  0000  PROC5:V      12.6
E1R:kΩ       0.0   CPU_T:°C      +33.12
E2R:kΩ       0.0
  
```

(POS. 12.6) *Disp. comm. Diagnostic values*

[Disp. comm. vars]

AL5

DCVLS

Create a list of diagnostic values on the instrument communication.

```

PPP link status → PPP_STATUS:  NETW  E_IP_HDR5:  3
MCPI socket status → MCPI_S: ESTABLISH E_IP_HDR6:  0
RxCNT: 1477149 E_IP_HDR7:  0
TxCNT: 6515456 E_IP_HDR8:  0
E_SR_LINK: 0 E_IP_HDR9:  0
E_PAKTLEN: 0 E_TCPHDR1:  0
E_NETLAYR: 0 E_TCPHDR2:  0
E_TSPLOYR: 0 E_UDPHDR1:  0
E_ARPHDR1: 0 E_UDPHDR2:  0
E_ARPHDR2: 0 E_UDPHDR3:  0
E_IP_HDR1: 0 E_ICMPHDR:  0
E_IP_HDR2: 0
E_IP_HDR3: 0
E_IP_HDR4: 0
  
```

Following are the states for the PPP link and MCPI to connect the device.

PPP link status:

- "UNDT" = undetermined
- "DEAD" = dead, link down, persistent condition
- "LCP" = LCP phase, transition condition
- "AUTH" = Authentication phase, transition condition
- "IPCP" = IP and DNS addressess assign phase, transition condition
- "NETW" = network established (normal persistent condition when the link is UP)
- "TERM" = link termination request, transition condition

MCPI link status:

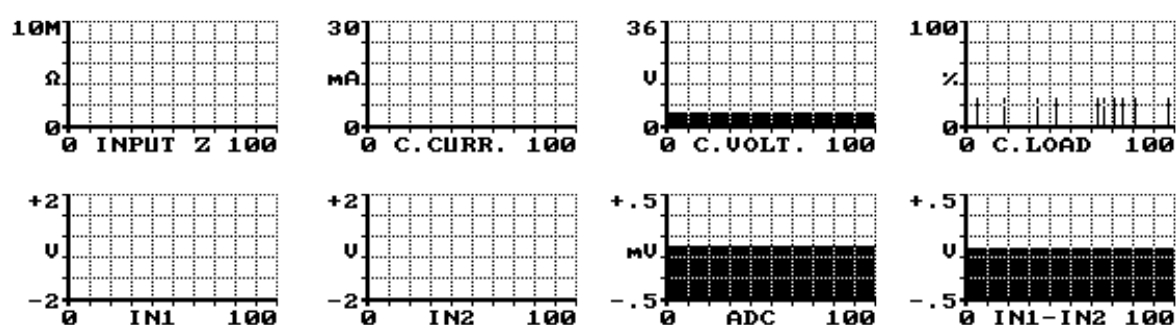
- "CLOSED" = socket closed
- "ACCEPT" = socket awaiting for new connection
- "ESTABLISH" = link established
- "CLS_WAIT" = waiting for closure
- "LAST_ACK" = lask ACK sent
- "FIN_WAIT" = (see TCP/IP RFC documentation)
- "TIME_WAIT" = (see TCP/IP RFC documentation)

(POS. 12.7) Display graphs

[Display graphs]

AL5 NO MCP COMMAND

This function displays graphs of input Z, C. current, C. Volt, C.Load, Input 1, Input 2, Input1-Input 2, Analog to Digital Converter.



(POS. 12.8) Generic sensor set

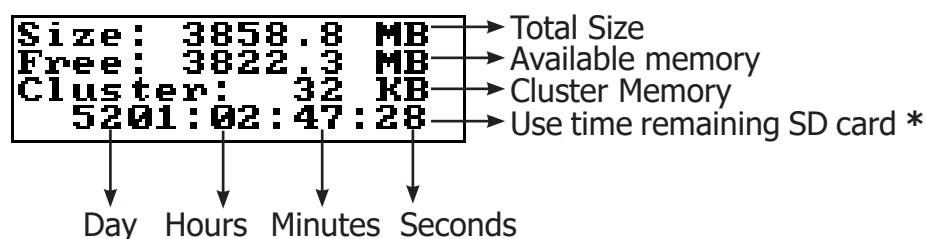
[Gen.sens. set]

AL5 NO MCP COMMAND

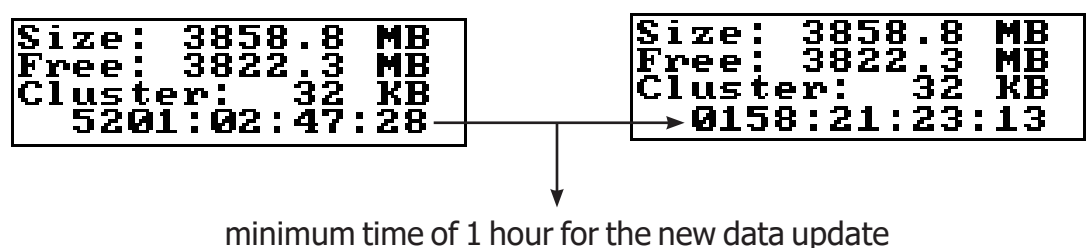
Automatic finding of a parameter set for a generic sensor.

(POS. 12.9) SD card status info. [SD card info] *ALO* **SDSTA**

This function shows the status of the SD card as shown in the following figure.



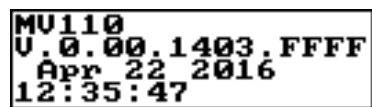
The statistical calculation is performed every time the command SDSTA is invoked. The statistical data are updated AUTOMATICALLY every day (24 hours) or manually each time the function is called to display or send the SDSTA command. However, given that for the detection is necessary a minimum of one hour, the data will not be recomputed before this time interval.



The SD card must be replaced only by the service. The use of commercial cards could disable some functionality of the instrument.

(POS. 12.10) Firmware info [Firmware info] *ALO* **MODSV**

Firmware info version/revision



(POS. 12.11) Board Serial Number [S/N= xxxxxx] *ALO* [SRNUM?]

View Board serial number. (read only)

(POS. 12.12) Working Time [WT= xxxx: xx: xx: xx] *ALO* [TWKTM?]

View Total working time instrument. (read only)

MENU 13 - SYSTEM

(POS. 13.1) Daylight saving time [Dayl. Saving= ON] *AL2* [DYSTE]

Daylight saving time change.

(POS. 13.2) Time zone [Time zone=h+xx.xx] *AL2* [TZONE]

Set time for geographic area

(POS. 13.3) Date and Time [xxxx/xx/xx-xx:xx] *AL2* [DTIME]

Set to system date and time

(POS. 13.4-5-6-7-8-9) Access level n° code [Ln xxxxxxxx] --- [L1ACD]-> [L6ACD]

This function enables or disables, for each access level code, the main menu functions.

Each level unlocks the functionality of the lower level. (Function 13.10 see page 31)

L1 code= ***** Access level value code 1 L4 code= ***** Access level value code 4

L2 code= ***** Access level value code 2 L5 code= ***** Access level value code 5

L3 code= ***** Access level value code 3 L6 code= ***** Access level value code 6

(POS. 13.10) Restricted access level [Restr. access= ON] *AL6* [RSARE]

Enable Or disable access level code. If active displays only the functions related to the level entered access.

IP ADDRESS SETTING (13.11-12-13)

(POS. 13.11) Device IP address [XXX.XXX.XXX.XXX] *AL3* [DIPAD]

Device IP network edress

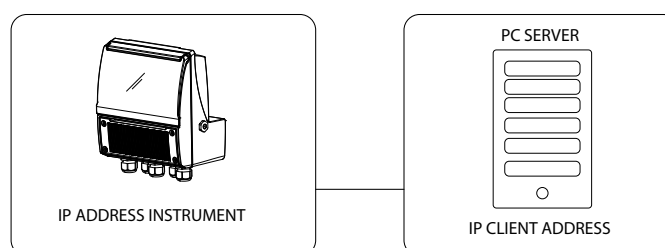
(POS. 13.12) Client IP address [XXX.XXX.XXX.XXX] *AL3* [CIPAD]

Client IP network adress

(POS. 13.13) Network mask [XXX.XXX.XXX.XXX] *AL3* [NETMS]

Network mask.

Caution: Changes to the functions of the points 13.11-13.12-13.13 are enabled after the drive device restart (see function 12.1 Self test for restart converter).



(POS. 13.14) Coefficient KT [KF=X.XXXXXX] *AL6* [CFFKT]

Gain correction coefficient (calculated automatically)

(POS. 13.15) Coefficient KS [KF=X.XXXXXX] *AL5* [CFFKS]

Correction coefficient constant instrumental

(POS. 13.16) Coefficient KR

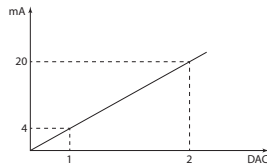
[KR=X.XXXXX]

AL5

[CFFKR]

Correction coefficient constant instrumental

DIGITAL ANALOG CONVERTER (Correction Parameters)(13.17-18-19-20)



The diagram shows how the DAC4-20mA parameters are setup. The DAC1 value corresponds to 4 mA corresponding to a zero flow rate, while the value of 20mA corresponds to a 100% of the flow rate.

(POS. 13.17) DAC1 4mA

[DAC1 4mA =XXXXX]

AL5

[C1CP1]

DAC1 out 4mA calibration point. (current output1 calibration point 1)

(POS. 13.18) DAC1 20mA

[DAC1 20mA=XXXXX]

AL5

[C1CP2]

DAC1 out 20mA calibration point. (current output1 calibration point 2)

(POS. 13.19) DAC2 4mA

[DAC1 20mA=XXXXX]

AL5

[C2CP1]

DAC2 out 4mA calibration point. (current output2 calibration point 1)

(POS. 13.20) DAC2 20mA

[DAC2 20mA=XXXXX]

AL5

[C2CP2]

DAC2 out 20mA calibration point (current output2 calibration point 2)

(POS. 13.21) Stand-BY

[STAND-BY]

AL3

[SSTBY]

Enable the converter standby state. It is enable by selecting chargeable battery in hw confing.

(POS. 13.22) Firmware Update

[FW update]

AL4

[FWUPD]

Enable firmware update. The firmware can be upload to the SD card (name.file).
MCP interface is activated by the command FWUPD = name.file

MENU 13 - SYSTEM: ONLY MCP FUNCTIONS

Unique Identity KEY	[MCP ONLY]	AL0	[UIKEY]
----------------------------	-------------------	------------	----------------

Device Unique Identity key

HardWare SET	[MCP ONLY]	AL0	[HWSET]
---------------------	-------------------	------------	----------------

Device hardware configuration

HardWare CODE	[MCP ONLY]	AL0	[HWCOD]
----------------------	-------------------	------------	----------------

Device hardware code

Calibr. Exec. status Memory	[MCP ONLY]	AL6	[CALXM]
------------------------------------	-------------------	------------	----------------

Calibration Execution status Memory. This function checks the instrument's internal calibration status.
CALXM=1 instrument calibrated correctly
CALXM=0 Invalid calibration / Calibration not completed (invalid calibration (if the function is zero, start the calibration function, MCP CALIC command)).

RTC Real time clock	[MCP ONLY]	AL2	[RTCAC]
----------------------------	-------------------	------------	----------------

RTC is used to set a correction factor for the internal clock.

For setting the correction date and time with MCP press 

Function CODE Selection	[MCP ONLY]	AL0	[FCODS]
--------------------------------	-------------------	------------	----------------

Select the function code

FuNction Enable State Selection	[MCP ONLY]	AL6	[FNESS]
--	-------------------	------------	----------------

Select the enable state of function

All FuNctions State Selection	[MCP ONLY]	AL6	[AFNSS]
--------------------------------------	-------------------	------------	----------------

Select enable state of ALL function

Quick Start FuNction Selection	[MCP ONLY]	AL6	[QSFNS]
---------------------------------------	-------------------	------------	----------------

Select function for quick start menu

Quick Start All Functions Selection	[MCP ONLY]	AL6	[QSAFS]
--	-------------------	------------	----------------

Select ALL function converter for quick start menu.

Quick start function Status LiST	[MCP ONLY]	AL6	[QSLST]
---	-------------------	------------	----------------

List quick start group functions

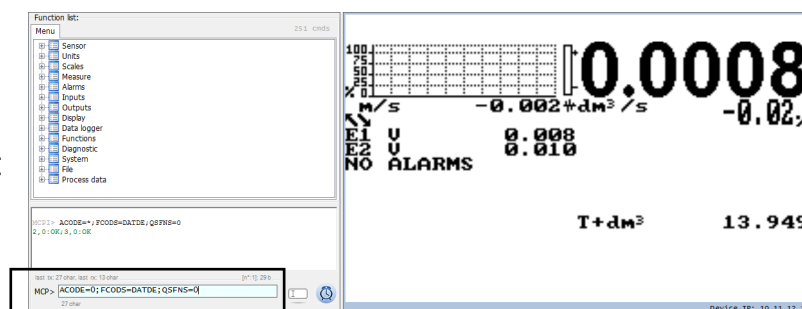
Function enable Status LiST	[MCP ONLY]	AL6	[FSLST]
------------------------------------	-------------------	------------	----------------

List enable status of functions

Access CODE	[MCP ONLY]	AL0	[ACODE]
-------------	------------	-----	---------

Input the right access code

Exemple set quick start menù function for mcp.



ACODE=0; FCODS=[MCP COMMAND]; QSFNS=1

Access Code INSERT MCP Command of the function to be activated in the QS menu. Quick Start Function Selection

LINK Terminate	[MCP ONLY]	AL0	[LTERM]
----------------	------------	-----	---------

Terminate the PPP data link

MCPI session QUIT	[MCP ONLY]	AL0	[MQUIT]
-------------------	------------	-----	---------

Quit the MCPI connection

Functions LIST	[MCP ONLY]	AL0	[FLIST]
----------------	------------	-----	---------

View list of all available converter functions.

Functions LIST Compact	[MCP ONLY]	AL0	[FLISC]
------------------------	------------	-----	---------

View compact list of all available converter functions.

Functions Menu SElection	[MCP ONLY]	AL0	[FMSEL]
--------------------------	------------	-----	---------

Select menu for functions list

ConFfiguration LiST	[MCP ONLY]	AL0	[CFLST]
---------------------	------------	-----	---------

Configuration parameter list. The list with the status / values of the converter parameter.

Volume Totalizer Total Positive Set	[MCP ONLY]	AL4	[VTTPS]
-------------------------------------	------------	-----	---------

Totaliz.T+ value set

Volume Totalizer Partial Positive Set	[MCP ONLY]	AL4	[VTPPS]
---------------------------------------	------------	-----	---------

Totaliz.P+ value set

<i>Volume Totalizer Total Negative Set</i>	[MCP ONLY]	<i>AL4</i>	[VTTNS]
Totaliz.T- value set			
<i>Volume Totalizer Partial Negative Set</i>	[MCP ONLY]	<i>AL4</i>	[VTPNS]
Totaliz.P- value set			
<i>Volume Total Positive Overflow Set</i>	[MCP ONLY]	<i>AL4</i>	[VTPOS]
Totaliz.T+ overflow value set			
<i>Volume Partial Positive Overflow Set</i>	[MCP ONLY]	<i>AL4</i>	[VPPOS]
Totaliz.P+ overflow value set			
<i>Volume Total Negative Overflow Set</i>	[MCP ONLY]	<i>AL4</i>	[VTNOS]
Totaliz.T- overflow value set			
<i>Volume Partial Negative Overflow Set</i>	[MCP ONLY]	<i>AL4</i>	[VPNOS]
Totaliz.P- overflow value set			
<i>CPU MaX.recorded temperature</i>	[MCP ONLY]	<i>AL6</i>	[CPUMX]
CPU max.recorded temperature			
<i>CPU MiN.recorded temperature</i>	[MCP ONLY]	<i>AL6</i>	[CPUMN]
CPU min.recorded temperature			
<i>Board T1 MaX.recorded temperature</i>	[MCP ONLY]	<i>AL6</i>	[BT1MX]
T1 max.recorded temperature			
<i>Board T1 MiN.recorded temperature</i>	[MCP ONLY]	<i>AL6</i>	[BT1MN]
T1 min.recorded temperature			
<i>Board T2 MaX.recorded temperature</i>	[MCP ONLY]	<i>AL6</i>	[BT2MX]
T2 max.recorded temperature			
<i>Board T2 MiN.recorded temperature</i>	[MCP ONLY]	<i>AL6</i>	[BT2MN]
T2 min.recorded temperature			
<i>Calibration OFset Register 0</i>	[MCP ONLY]	<i>AL6</i>	[COFR0]
Calibration offset register 0			
<i>Calibration OFset Register 1</i>	[MCP ONLY]	<i>AL6</i>	[COFR1]
Calibration offset register 1			

<i>Calibration GAIN Register 0</i>	[MCP ONLY]	<i>AL6</i>	[CGAR0]
---	-------------------	-------------------	----------------

Calibration gain register 0

<i>Calibration GAIN Register 1</i>	[MCP ONLY]	<i>AL6</i>	[CGAR1]
---	-------------------	-------------------	----------------

Calibration gain register 1

<i>Calibration GAIN Register 2</i>	[MCP ONLY]	<i>AL6</i>	[CGAR2]
---	-------------------	-------------------	----------------

Calibration gain register 2

<i>Calibration GAIN Register 3</i>	[MCP ONLY]	<i>AL6</i>	[CGAR3]
---	-------------------	-------------------	----------------

Calibration gain register 3

<i>Calibration GAIN Register C</i>	[MCP ONLY]	<i>AL6</i>	[CGARC]
---	-------------------	-------------------	----------------

Calibration gain register C

MENU 14 - FILE (ONLY MCP)

<i>File Transfer ABoRt</i>	[MCP ONLY]	AL2	[FTABR]
Abort the current File Transfer			
<i>File Transfer STAtE</i>	[MCP ONLY]	AL0	[FTSTA]
Show the File Transfer state			
<i>Read Last EVenTs</i>	[MCP ONLY]	AL2	[RLEVT]
Read the latest system events			
<i>Read All EVenTs</i>	[MCP ONLY]	AL2	[RAEVT]
Read all current system events			
<i>Read Last Logged DaTa</i>	[MCP ONLY]	AL2	[RLLDT]
Read the latest logged data			
<i>Read All Logged DaTa</i>	[MCP ONLY]	AL2	[RLLDT]
Read all current logged data			
<i>Read Last Sensor Verify Data</i>	[MCP ONLY]	AL2	[RLSVD]
Read the latest sensor ver. data			
<i>Read All Sensor Verify Data</i>	[MCP ONLY]	AL2	[RASVD]
Read all sensor verify data			
<i>File SEND</i>	[MCP ONLY]	AL2	[FSEND]
Set file name for read operation			
<i>File ReCeIvE</i>	[MCP ONLY]	AL5	[FRCVE]
Set file name for write operation			
<i>File ReCeive APpend mode</i>	[MCP ONLY]	AL5	[FRCAP]
Set file name for write-append			
<i>File OFFSet position</i>	[MCP ONLY]	AL2	[FOFFS]
Set file offset position			
<i>ConFiGuration file WRite</i>	[MCP ONLY]	AL2	[CFGWR]
Save the configuration to a file			
<i>ConFiGuration file ReaD</i>	[MCP ONLY]	AL2	[CFGRD]
Read the configuration from file			

<i>FuNction list file WRite</i>	[MCP ONLY]	<i>AL2</i>	[FNCWR]
--	-------------------	-------------------	----------------

Save the functions list to file

<i>Function Enable Status WRite</i>	[MCP ONLY]	<i>AL6</i>	[FESWR]
--	-------------------	-------------------	----------------

Save function enable status to file

<i>Quick Start function Status WRite</i>	[MCP ONLY]	<i>AL6</i>	[QSSWR]
---	-------------------	-------------------	----------------

Save quick start function enable.

MENU 15 - PROCESS DATA (ONLY MCP)

<i>OUTput 1 Set</i>	[MCP ONLY]	<i>AL0</i>	[OUT1S]
Set value for digital output 1			
<i>OUTput 2 Set</i>	[MCP ONLY]	<i>AL0</i>	[OUT2S]
Set value for digital output 2			
<i>Digital INput 1 Status</i>	[MCP ONLY]	<i>AL0</i>	[DIN1S]
Digital input 1 status read			
<i>Flow Rate Full Scale in chosen Units</i>	[MCP ONLY]	<i>AL0</i>	[FRFSU]
F.rate f.scale in chosen units			
<i>Flow Rate Value PerCentage</i>	[MCP ONLY]	<i>AL0</i>	[FRVPC]
Flow rate value in percentage			
<i>Flow Rate Value Percentage without cut-off</i>	[MCP ONLY]	<i>AL0</i>	[FRVPX]
F.rate in perc.without cut-off			
<i>Flow Rate Value Binary without cut-off</i>	[MCP ONLY]	<i>AL0</i>	[FRVBX]
F.rate in binary.without cut-off			
<i>Flow Rate Value Technical Unit</i>	[MCP ONLY]	<i>AL0</i>	[FRVTU]
F.rate value in unit of measure			
<i>Volume Totalizer Total Positive Value</i>	[MCP ONLY]	<i>AL0</i>	[VTTTPV]
Totaliz.T+ read value			
<i>Volume Totalizer Partial Positive Value</i>	[MCP ONLY]	<i>AL0</i>	[VTPPV]
Totaliz.P+ read value			
<i>Volume Totalizer Total Negative Value</i>	[MCP ONLY]	<i>AL0</i>	[VTTNV]
Totaliz.T- read value			
<i>Volume Totalizer Partial Negative Value</i>	[MCP ONLY]	<i>AL0</i>	[VTPNV]
Totaliz.P- read value			
<i>Volume Totalizer Total Positive Overflow</i>	[MCP ONLY]	<i>AL0</i>	[VTTPO]
Totaliz.T+ number of overflows			
<i>Volume Totalizer Partial Positive Overflow</i>	[MCP ONLY]	<i>AL0</i>	[VTPPO]
Totaliz.P+ number of overflows			
<i>Volume Totalizer Total Negative Overflow</i>	[MCP ONLY]	<i>AL0</i>	[VTTNO]
Totaliz.T- number of overflows			

<i>Volume Totalizer Partial Negative Overflow</i>	[MCP ONLY]	<i>AL0</i>	[VTPNO]
Totaliz.P- number of overflows			
<i>Board TeMPeratures</i>	[MCP ONLY]	<i>AL0</i>	[BTMPS]
Board temperatures			
<i>CPU temperature</i>	[MCP ONLY]	<i>AL0</i>	[CPUTP]
CPU temperature			
<i>Sensor CoiLs TemPerature</i>	[MCP ONLY]	<i>AL0</i>	[SCLTP]
sensor's coils temperature			
<i>LiQuid VELocity</i>	[MCP ONLY]	<i>AL0</i>	[LQVEL]
Liquid velocity			
<i>AVeraGe process data Samples Number</i>	[MCP ONLY]	<i>AL0</i>	[AVGSN]
N.of samples for averaged values			
<i>ALARM status</i>	[MCP ONLY]	<i>AL0</i>	[ALARM]
Active alarm(s) status			
<i>Sensor TeSt Result Code</i>	[MCP ONLY]	<i>AL0</i>	[STSRC]
Sensor test result code			
<i>Main power status</i>	[MCP ONLY]	<i>AL0</i>	[MPWRS]
Status of main power supply			
<i>INput RESistance</i>	[MCP ONLY]	<i>AL0</i>	[INRES]
Equivalent Input resistance			
<i>INput VoLtageS</i>	[MCP ONLY]	<i>AL0</i>	[INVLS]
Electrodes input voltages			
<i>System Battery Voltage</i>	[MCP ONLY]	<i>AL0</i>	[SBVLT?]
View battery voltage			
<i>System Battery Charge Status</i>	[MCP ONLY]	<i>AL0</i>	[SBCHS?]
View system battery charge status			
<i>Measure BUFFers</i>	[MCP ONLY]	<i>AL0</i>	[MBUFF]
Measure buffers data read			
<i>SEquence NumBer</i>	[MCP ONLY]	<i>AL0</i>	[SEQNB]
Sequence number			

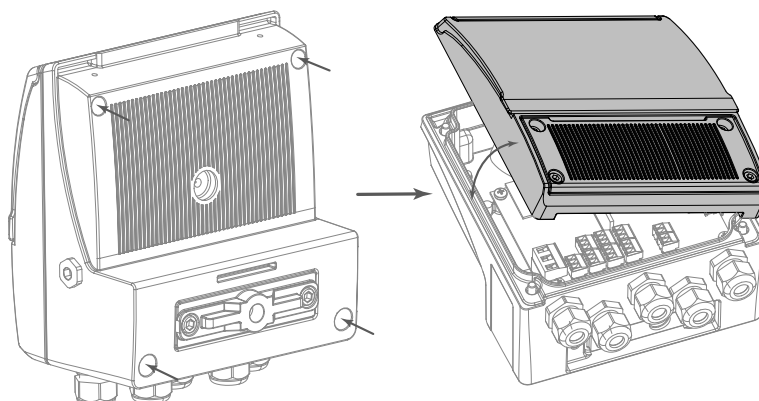
METER DATA

This function allows the import of data from one converter to another up to level 4 included. The hardware configurations and the corresponding calibration values are not restored. The "data import" procedure can be performed **one time only**, since the directory, according to the board's SERIAL NUMBER, will be renamed.

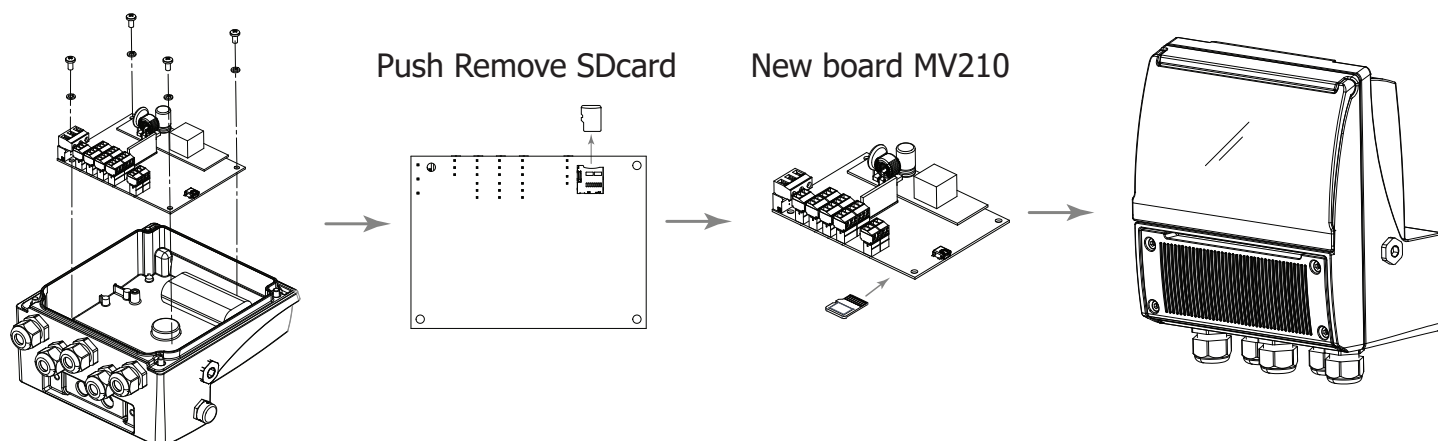
OPERATING PROCEDURE TO CHANGE THE CONVERTER BOARD



Remove the 4 screws (see "MV210 construction" page10) to remove the main housing. Attention of any electrical cables.



Remove the 4 screws (see "MV210 construction" page10) to remove the MV210 board. Attention of any electrical cables. Remove the SD card and insert the new MV210 board.



Assembled the card turn on the converter and enable [Save Conv. F. def.] function to restore data.



View function "11.8" page30.

B.I.V. (BUILT-IN VERIFICATOR)

BIV, abbreviation for Built In Verificator, is available as option for MV210 converters and must be enabled by the manufacturer. It is also necessary that the SD card is activated to store saved data. The analysis of collected data performed by a dedicated IsoBIV software running on another device (PC). The simplicity of test procedures minimize the risk of handling errors; maximum safety and reliability thanks to the traceable factory calibration and internal references complement the safety by design principle with minimal failure rates

IsoBIV allows to create and print a report as validation of device functionality/measure error.

Operation and Conditions of Use.

The system is based on periodic measurements performed every hour or using a manual command (MCP command = SVERC). The sensor parameters are measured and compared with previously measured and stored reference values. Each time the system performs a series of measurements on the sensor and records them in a file called "STESTLOG.CSV", which resides in the main directory of the SD memory of the converter.

The sensor test can also be carried out without the active BIV system, but in this case only the presence of isolation losses and the overall good functioning of the sensor such the coil resistance, the excitation current and the rising times of the current within the generic limits that guarantee operation. Instead if BIV is active, the measurements are deeper and the measured values are tested by comparing them with a set of characteristic sensor parameters measured at the time of installation.

Saving Reference Values (Characteristic Parameters)

After sensor installation, the parameters that will be used as reference for the BIV system and the IsoBIV data analysis software must to be measured.

The characteristic values of the coil circuits are saved in the converter memory at the factory before to ship the instrument. For the reference measure of the electrodes circuits, there is a specific function that perform the measures of voltage and resistance at the installation site. This function is managed ONLY by the IsoBIV program, which through a simple wizard will set the converter to perform the measures in the specific measurement point where the meter is installed.

To activate BIV, these functions must be verified:

- ☐ **[ASVFE=1]** : It enable the sensor's automatic test every hour. The ASVFE function in Menu 1 with access level 3 can be also activated using the instrument's display.

This feature can be enabled even if the SD card is not installed and if the BIV function is not active; in such a case the sensor file is NOT created and any alarms will be generated either if the data deviation from the reference data is outside the sesnor's limits.

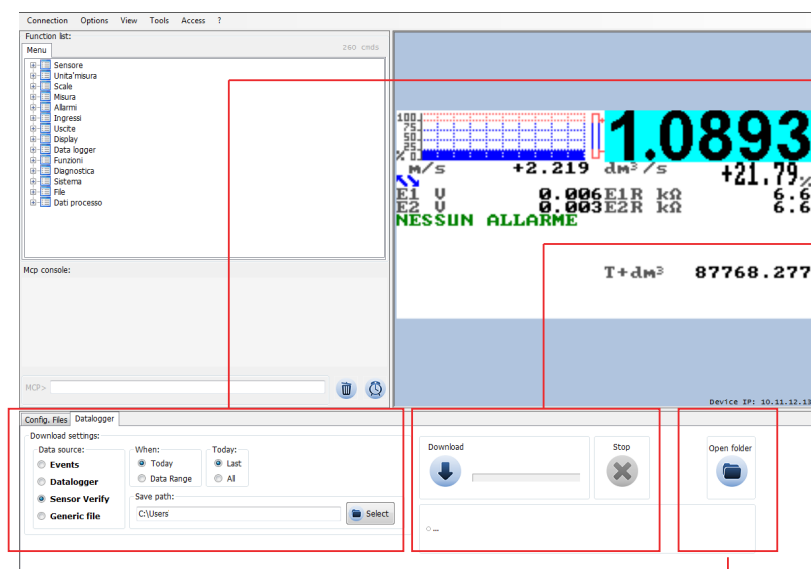
Practically, in the absence of the necessary hardware permissions, this function is useful to test the insulation of the coils.

Opening and reading manually the files STESTLOG.CSV

The list below describes the steps for saving and reading STESTLOG.CSV file.

❑ OPEN INTERFACE STARTING THE PROGRAM MCP 

❑ FOLLOW THE STEPS HERE BELOW



- 1
 - Select Sensor Verify
 - Choose the date or reference period of the which one wants to carry out the evaluation of the
 - Choose the directory where to save the file

- 2
 - Press Download to download the file.
 - Press Download to download the file.
 - At the end of the download the stestlog.csv file will be automatically opened by the program

Search for the consultation of the files inside of the drive SD card.

Path structure for searching the files in the SD card of the converter

